Floor Plan Parsing Without Learned Models: Walls, Openings and Room Graphs

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***Abstract*—We present a rule-based parser that converts a raster architectural floor plan into walls, doors, windows, rooms and a room–door graph using only classical image processing: thresholding, mathematical morphology, the progressive probabilistic Hough transform, template matching and the watershed transform. No learned model is used at any stage. Walls are extracted at two morphological scales and vectorized from their skeleton; openings are found as gaps between wall ends and classified by fitting quarter-circle door arcs and parallel window lines; rooms are segmented by closing the openings and splitting the free space with a distance-transform watershed. On the solid-wall subset of the CVC-FP dataset, measured once on a held-out test split of 24 plans, the method reaches a wall pixel IoU of 0.913, door F1 0.904, window F1 0.830 and room F1 0.770 under strict one-to-one matching at IoU ≥ 0.5. All thresholds were chosen on a separate development split of 62 plans. We compare against published results on the same data, analyze the remaining errors, and report the drawing styles where the approach fails.**

***Index Terms*—floor plan analysis, document image analysis, mathematical morphology, Hough transform, template matching, watershed segmentation**

I. Introduction

Architectural floor plans are routinely exchanged as raster images in which the semantic structure (which strokes are walls, where the doors are, which spaces are rooms) is lost. Recovering that structure is a classical problem of graphics recognition [1]–[3]. Recent work relies mostly on trained detectors and segmentation networks [4]–[7]. This project deliberately does not: the task, set for an image processing course, is to solve the problem with classical operators only, so that every decision of the system can be traced to a measurable image property.

The contributions of this work are:

- a complete pipeline from pixels to a room–door graph built only from thresholding, morphology, Hough lines, template matching and the watershed transform;
- geometric tests for openings: gap candidates from wall ends and mask corners, a four-hypothesis quarter-arc template for doors, and line- and arc-based tests for single and double casement windows;
- an evaluation protocol with strict one-to-one matching for every element type, a development/test split fixed in advance, and a single test measurement, together with an error analysis that separates fixable failures from failures the approach cannot address.

II. Related Work

Macé et al. [1] detect rooms by separating thick and thin lines, finding walls and doors with Hough-based primitives, and decomposing the remaining space. Ahmed et al. [2] extend this with a three-way thick/medium/thin separation and a building boundary, and later add text labels from OCR to split rooms that have no physical separation [8]. Their evaluation uses the protocol of Phillips and Chhabra [9], which gives partial credit for one-to-many and many-to-one matches.

De las Heras et al. introduced patch-based wall segmentation [10], a full statistical and structural pipeline [3], and the CVC-FP dataset [11] used here. Dodge et al. [4] compare thresholding, patch classifiers and fully convolutional networks for wall segmentation on CVC-FP. Swaileh et al. [12] generate wall masks for historical plans with connected-component filtering and steerable filters before training a U-Net. Cabrera Vargas [13] studies walls and rooms in multi-unit plans, and Rezvanifar [14] spots symbols, including door arcs found by partial-ellipse voting. FloorYOLO [7] detects doors and windows on CVC-FP with an attention-augmented YOLOv8n.

We did not find a published system that extracts walls, doors, windows, rooms and the room–door graph on CVC-FP without any learned component and reports all of them; the closest is [3], whose full text we could not access.

III. Data and Evaluation

A. Dataset and split

CVC-FP [11] contains 122 annotated plans in four drawing styles. Walls, rooms, doors, windows, wall-less separations and parking (garage) objects are annotated as polygons, and rooms are linked to their doors by *access* relations. The largest group (90 plans, 2479×3508 px) draws walls as solid black bands in 86 plans; the other styles use hatched fills or double outlines. The method targets solid walls; we report the solid-wall subset as the main result and all styles separately.

There is no official split. We hold out 30% of each style group by the SHA-1 order of the plan names: 86 development and 36 test plans (62 and 24 solid-wall plans). Every threshold and design choice was made on the development plans; the test plans were evaluated once, after the method was frozen.

B. Metrics

Walls: pixel IoU between the predicted wall mask and the union of the ground-truth wall polygons, averaged over plans.

Doors and windows: each predicted polygon is matched to at most one ground-truth polygon, greedily by decreasing IoU, and counts as correct if the IoU is at least 0.5; we also report the same matching on axis-aligned bounding boxes for comparison with object detectors. *Rooms:* the same one-to-one rule between predicted room regions and ground-truth room polygons. *Graph edges:* a predicted room–door–room edge is correct if its door matches a ground-truth door, both rooms match ground-truth rooms, and the ground truth links that door to exactly those two rooms. Precision (P), recall (R) and F1 are computed from the pooled counts.

These metrics are strict. Merging two rooms counts as one miss and one false region, with no partial credit, unlike the DR/RA protocol used in [2], [8].

IV. Method

Fig. 1 shows the stages on a crop of a development plan.

A. Preprocessing

The grayscale image is binarized with Otsu's threshold [15]. Enclosed paper regions of at most 4 pixels, which occur inside scanned wall bands, are filled.

B. Two-scale wall mask

A single morphological opening [16] cannot separate thin partition walls (about 7–9 px) from strokes of the same width such as stair treads. We therefore use two scales. The *thick* mask is the opening with a 13×13 square and is accepted as wall directly. The *thin* candidates are the opening with a 7×7 square minus the dilated thick mask; they are vectorized (Section IV-C) and a thin segment is accepted as a wall only if it is at least 30 px long, its median transverse width is at most 10 px, its width varies little (interquartile range at most 0.2 of the median), and it has at most one parallel neighbor within 60 px (stair treads come in parallel series).

A square structuring element is direction dependent: a 7×7 square fits into a 45° band only if the band is about $7\sqrt{2}$ px wide, so thin diagonal walls disappear. The pixels kept by an opening with a 7 px disk but not by the square opening form a separate mask; segments from that mask that are at least 10° off both axes and pass the same thin-wall rules are added. Keeping the square candidates unchanged and only adding diagonal segments matters: merging the disk opening into the candidate mask shifted existing segments and changed downstream results.

C. Vectorization

Each mask is thinned [17] and line segments are found with the progressive probabilistic Hough transform [18] (accumulator threshold 10, minimum length 15 px, maximum gap 20 px). Collinear segments (angle within 3°, offset within 3 px, gap within 20 px) are merged; each segment is extended along its direction to the mask boundary, its thickness is measured as the median transverse extent at quarter-pixel steps, and end points near another wall (angle at least 20°) are moved to the intersection of the two center lines. The final wall mask is the thick mask plus the rectangles of the accepted thin segments.

D. Gap candidates

Openings appear as interruptions of wall lines. From each segment end a ray is cast along the segment; if it leaves the wall and meets wall again after 40–200 px, the free interval is a candidate gap. At an end that is joined to another wall the ray first crosses that wall, and the wall it meets must be collinear (within 10°); otherwise the ray would cross the room from a T-junction. Further candidates come from rays along both faces of every approximately right-angled convex corner of the wall mask, which covers short or slanted wall ends that produced no segment. Gaps of 150–450 px are searched separately for wide windows; there the opposite wall must be collinear, or be a short pier whose local transverse width matches the source wall.

E. Doors

A door is drawn as a leaf and a quarter-circle arc centered at the hinge. For a gap of width g we test four hypotheses (hinge at either end, opening to either side), with the hinge on the wall face or on its center line, and radii $r \in \{0.85, 0.92, 1.0, 1.06\}\, g$; for door-size arcs ($r \geq 75$ px) the radii $0.83g$ to $1.07g$ in steps of $0.04g$ are added. For each hypothesis 24 points of the quarter arc between 8° and 82° are sampled; the score is the fraction of points within 3 px of thin ink, i.e. the binary image with the dilated wall mask removed. A gap is a door if the best score is at least 0.8 and $r \geq 75$ px; smaller arcs are casement windows. Overlapping detections are suppressed by the IoU of their swing sectors.

The radius step matters. With a step of about 7 px, twice the point tolerance, the correct radius can fall between two tested values (Fig. 2); adding the intermediate radii for door-size arcs found 32 more doors on the development set with an unchanged window F1. Applying them to the smaller casement-window arcs as well turned windows drawn with parallel lines into casement detections and lowered window F1, so they are limited to door-size arcs. Candidates from junction ends that hit a perpendicular wall face (doors at the foot of a T-junction) are tested for arcs only.

F. Windows

In a gap without a door, a window is reported if at least two parallel ink runs, each covering at least 80% of the gap, lie inside the wall band. A double casement window has two leaves hinged at both ends of the gap that meet in the middle; we test two arcs of radius $g/2$ and require both to score at least 0.8. Following the ground-truth convention, the polygon of a casement window is the gap rectangle together with the swing sectors.

G. Rooms and graph

The closed mask is the wall mask plus 5 px beams across detected doors, windows and wide openings, across half-open doors (a partial arc plus a leaf at 30°, 45° or 60°, score at least 0.95, used for closing only) and across the remaining gap candidates. A beam is not drawn from a wall end that lies on an interrupted wall line, i.e. with wall on both sides within

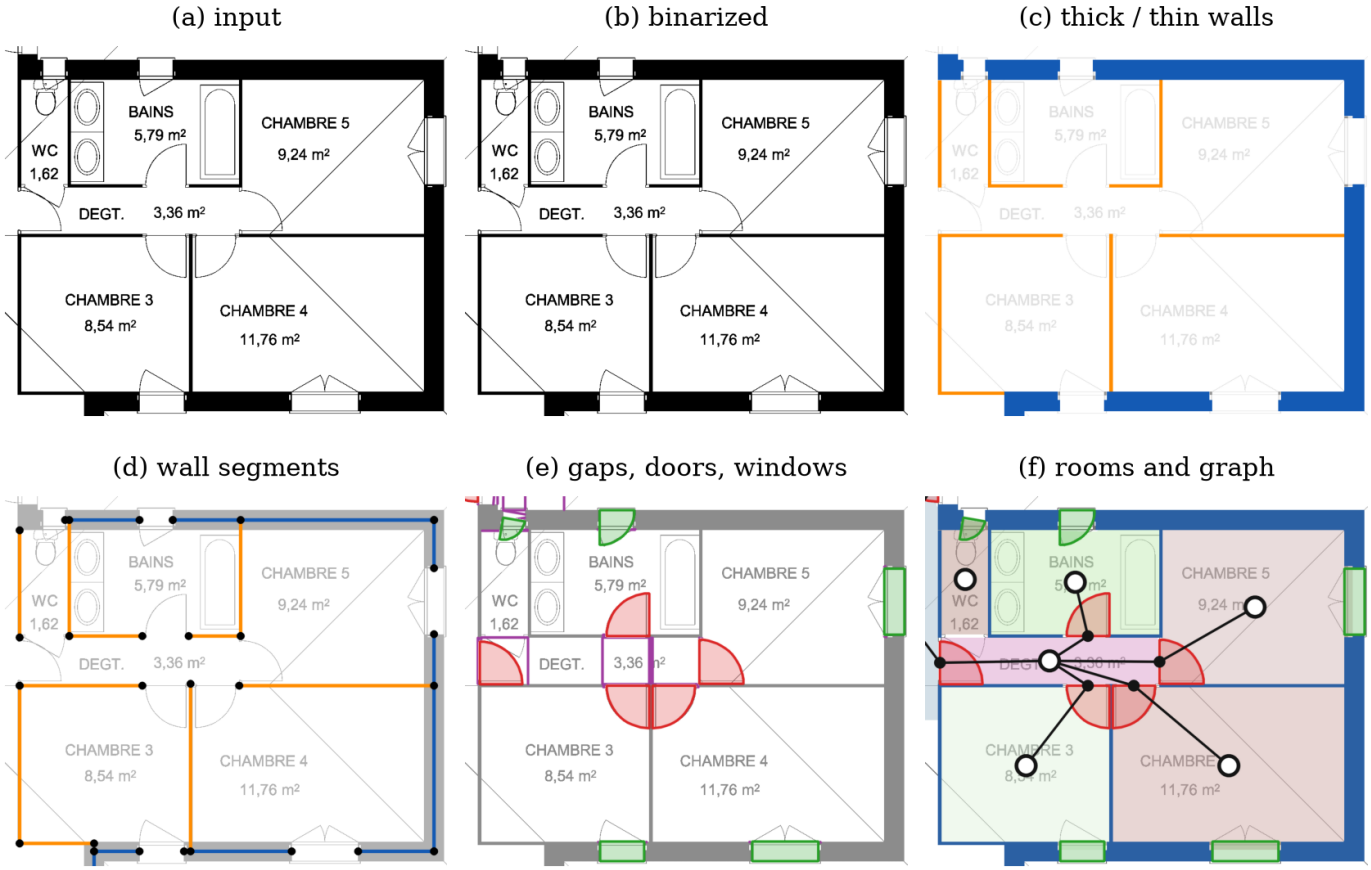


Fig. 1. Pipeline stages on a development plan. (c) Thick mask (blue), thin-scale candidates (gray) and thin walls accepted by the rules (orange). (d) Wall segments from the skeleton and Hough transform, with their end points. (e) Gap candidates (purple), detected doors (red) and windows (green). (f) Rooms and the room–door graph.

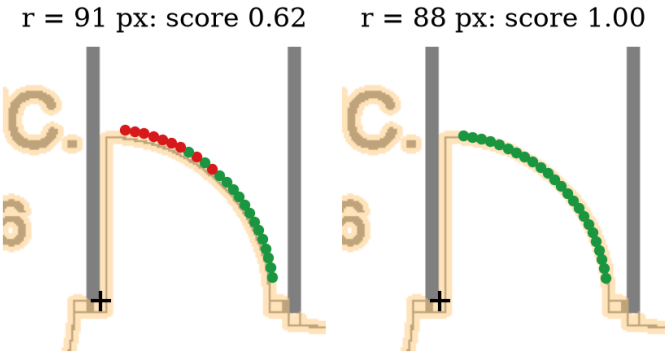


Fig. 2. Arc template on a development plan. Samples within 3 px of thin ink are green, others red. A radius of 91 px, from the original coarse radius scan, scores 0.62 and the door is missed; the true radius, 88 px, scores 1.00.

220 px: such ends are typically partitions meeting a corridor between two doors, and their rays cut the corridor.

The building footprint is the closing of the wall mask, with wide openings closed, by a 301×301 square, with holes filled. Regions inside the footprint whose boundary is less than 80% wall or beam are concave exterior spaces such as terraces. The free space is split by a watershed [19] on the distance transform, with markers where the distance to the nearest wall is at least 45 px (about half a door width); adjacent basins merge when the saddle between them is at least 0.8 of the smaller peak, so that L-shaped rooms stay whole while rooms joined by an unclosed doorway are separated. Regions of at least 5000 px are rooms. For every door, the labels a few pixels on both sides of the door line give a room–room or room–exterior edge in a multigraph.

V. Results

A. Test results

Table I gives the single test measurement next to the development results on which the method was tuned. On the solid-wall plans, wall, door, window and room scores differ by at most 0.015 between the two splits. The edge F1 is notably higher on the test split; it is computed from only 188 test edges and depends on whether the corridor, to which most doors connect, is found as one room, so we do not read the difference as an improvement. Fig. 3 shows the test plan with the median room F1.

B. Development history

Table II lists the main design steps and their effect on the development set. Each step was introduced after classifying

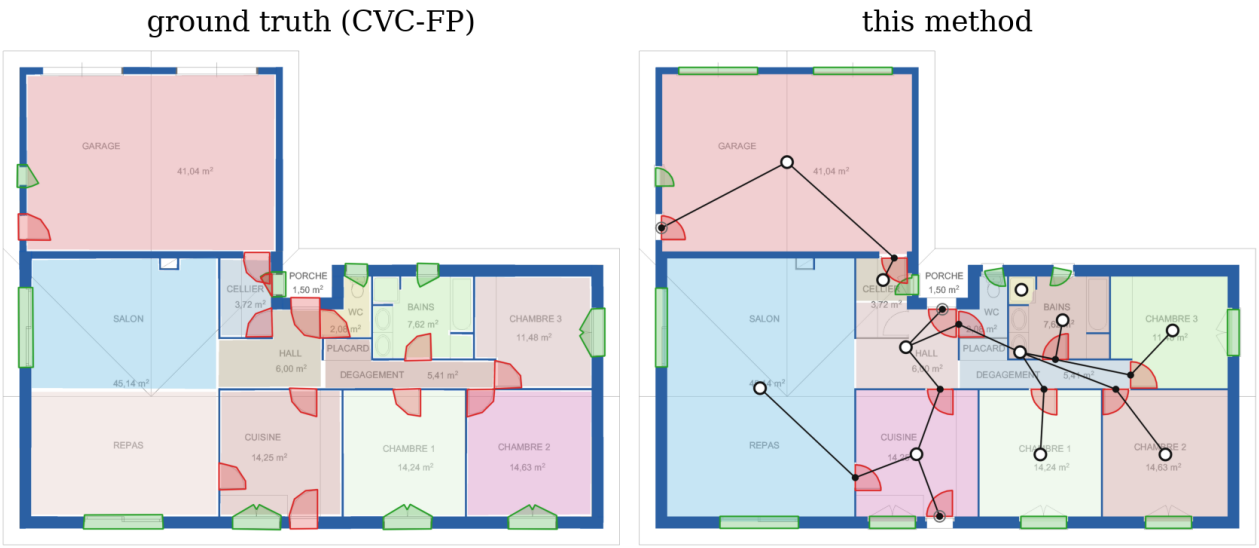


Fig. 3. Test plan with the median room F1 (0.78). Walls blue, doors red, windows green, rooms in pastel colors; right: the room–door graph. Salon and dining area are one room in our output: the ground truth separates them without a wall.

TABLE I
DEVELOPMENT AND TEST RESULTS, SOLID-WALL PLANS (P / R / F1; ONE-TO-ONE MATCHING AT IOU ≥ 0.5).

	development (62)	test (24)
wall pixel IoU	0.909	0.913
doors	0.957 / 0.836 / 0.892	0.985 / 0.835 / 0.904
windows	0.882 / 0.798 / 0.838	0.863 / 0.798 / 0.830
rooms	0.859 / 0.719 / 0.783	0.825 / 0.722 / 0.770
room–door edges	0.446 / 0.350 / 0.392	0.580 / 0.441 / 0.502
all styles (86 / 36 plans), F1		
doors	0.730	0.770
windows	0.652	0.625
rooms	0.647	0.651

TABLE II
EFFECT OF DESIGN STEPS ON THE DEVELOPMENT SET (F1).

step	F1
walls (median pixel IoU)	
single opening	0.84
two scales with thin-wall rules	0.926
windows	
parallel lines and small arcs	0.671
+ double casement test	0.771
+ wide gaps, short piers	0.839
doors	
arc score ≥ 0.8, $r \geq 75$ px	0.854
+ finer radius scan, T-junction gaps	0.885
+ diagonal thin walls	0.892
rooms	
connected components	0.534
+ wide openings, footprint wall share	0.626
+ watershed splitting	0.706
+ beam filter, half-open doors	0.734
+ door and wall improvements	0.783

the remaining errors of the previous version; the classification scripts are part of the repository.

C. Comparison with published results

Table III collects published numbers on CVC-FP. None of them uses our protocol and the comparison must be read with the differences in mind. Dodge et al. report the wall Jaccard index on all 122 plans with five-fold cross-validation; our all-style mean IoU (0.634 on the test split) lies close to their thresholding baselines and well below the learned models, because the method finds almost no walls in hatched styles. On solid walls our 0.913 exceeds the only classical mask generation we found, 0.831 by Swaileh et al., whose input had a textured background added. FloorYOLO reports precision and recall on a 25-plan validation split of all styles at 640×640 input; our door recall is higher on solid-wall plans and lower over all styles, and our window detection is below theirs. The room numbers of Macé et al. and Ahmed et al. give partial credit to split and merged rooms and are not directly comparable to our strict one-to-one recall and precision.

D. Runtime

The pipeline runs on a CPU in a median of 6–7 s per plan of 2479×3508 px (Python, OpenCV, scikit-image), with arc tests and watershed segmentation taking most of the time.

VI. Error Analysis and Limitations

Fig. 4 shows the main failure types. In our error analysis on the development set, 93 of the 602 ground-truth rooms were merged with a neighbor from which the ground truth separates them without a wall (Fig. 4a); no geometric cue exists in the drawing, which caps room recall at about 0.85 for this approach. Ahmed et al. [8] address this with OCR of room labels. Most other room merges occur where the door between two rooms was not found; the remaining missed doors

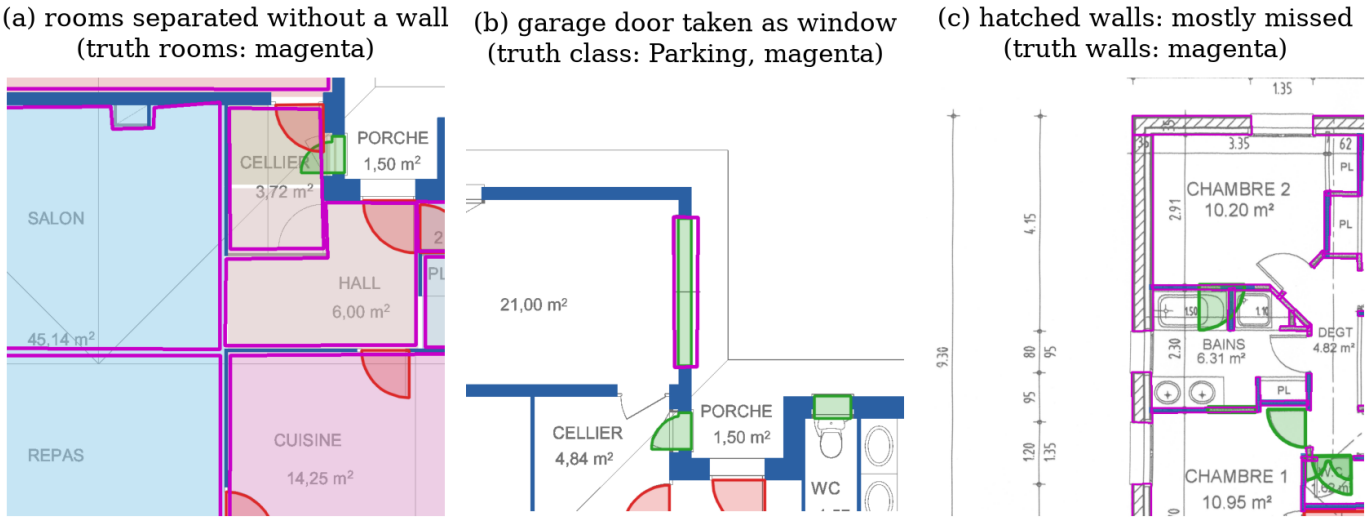


Fig. 4. Failure types. (a) Two ground-truth rooms separated without a wall are one region (test plan). (b) A garage door, annotated as a separate *parking* object, is detected as a wide window. (c) Hatched walls (another drawing style) are mostly missed.

TABLE III
PUBLISHED RESULTS ON CVC-FP AND OURS (TEST SPLIT).

work	setting	value
walls, Jaccard index / IoU		
Dodge [4]	Otsu / + morphology	0.415 / 0.619
	patches + RF / FCN-2s	0.792 / 0.892
Swaileh [12]	classical, solid walls	0.831
ours	all styles, mean / pooled	0.634 / 0.411
ours	solid walls	0.913
doors, box P / R		
FloorYOLO [7]	all styles	0.810 / 0.786
ours	all styles	0.966 / 0.640
ours	solid walls	0.985 / 0.835
windows, box P / R		
FloorYOLO [7]	all styles	0.791 / 0.727
ours	solid walls	0.671 / 0.621
rooms (DR / RA, partial credit)		
Macé [1]*	80 plans	0.85 / 0.69
Ahmed [2]	same 80 plans	0.89 / 0.79
Ahmed [8]	+ OCR room labels	0.85 / 0.82
rooms (R / P, strict one-to-one)		
ours	solid walls	0.722 / 0.825

*as reported in [2].

are mostly drawn half open or lie in walls that the wall step missed. Garage doors, which CVC-FP annotates as a separate *parking* class, are detected as wide windows (Fig. 4b). Stair base lines that are as thick as partition walls and touch no wall are accepted as walls; rules to reject them removed more true walls than false ones.

The thresholds are in pixels and tied to the scan resolution of CVC-FP; a different scale would need them rescaled, for example from the median wall thickness. Hatched and double-outline wall styles are outside the scope (Fig. 4c): the morphological wall model assumes solid bands.

VII. Conclusion

A pipeline built only from classical operators recovers walls, doors, windows and rooms of solid-wall floor plans with test F1 scores between 0.77 and 0.90 and a wall IoU above 0.91, and its errors can be traced to specific geometric causes. The largest remaining error, rooms separated without a wall, cannot be solved from the drawing geometry alone; reading room labels and normalizing the pixel thresholds by the measured wall thickness are the natural next steps.

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